Lesson5 Wheel and Axle

Objective: You will be able to

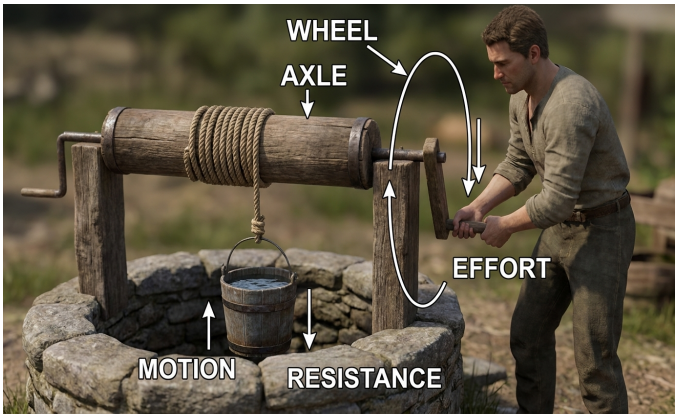
-Describe parts of wheel and axle

- Calculate mechanical advantage.

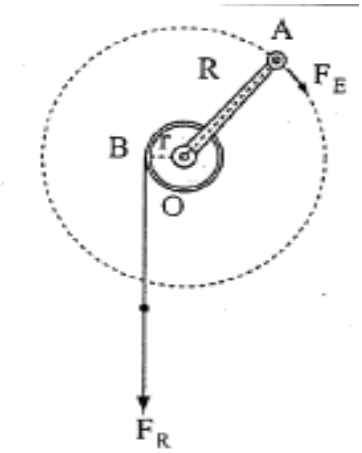
-Use wheel and axle in daily life.

1.Wheel and Axle

A wheel and axle is a lever with wheel which can rotate about a fixed axis.





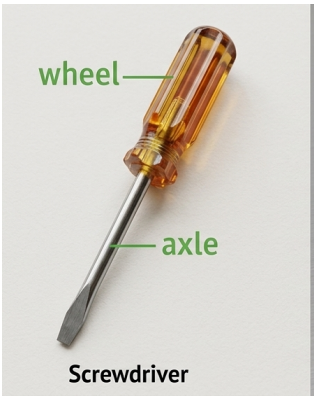


In order for the wheel and axle to be in equilibrium, the moment of load and moment of effort are equal.

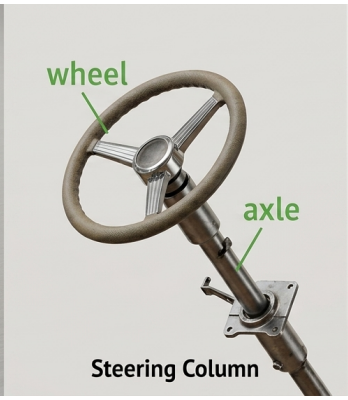
$$F_E \times R = F_R \times r$$



The pictures below show more wheel and axle.









2.Mechanical advantage is the ratio of output force(load) to input force(effort).

$$MA = \frac{F_R}{F_E} = \frac{R}{r}$$

Practice

1.A cylinder of a wheel and axle having a diameter of 40cm is used to lift a crate of 150N. The force applied to the handle is only 100N, Calculate the length of handle.

2.A wheel of diameter 14cm is fixed with a handle which make a circular path of diameter of 70cm. The weight of 200N is suspended to the wheel.

A. Find the moment of 200N.

B. Calculate the effort at the handle to balance the weight.